

BHUPESH DANEWA

Bikaner, Rajasthan

+91-9468601702 prince.danewa@gmail.com [linkedin.com/in/bhupesh-danewa](https://www.linkedin.com/in/bhupesh-danewa) github.com/bhupeshdanewa07

Education

Maulana Azad National Institute of Technology

M.Tech in Artificial Intelligence

2023 – 2025

Bhopal, Madhya Pradesh

Govt. Engineering College

B.Tech in Civil Engineering

2017 – 2021

Bikaner, Rajasthan

Work Experience

National Institute of Electronics & Information Technology

Sep 2025 – Present

Faculty

Bikaner

- Delivered instructions in Data Structures, Algorithms, Python, Web Design, AI Foundations, and Machine Learning at NIELIT Bikaner, ensuring curriculum alignment with industry standards and competency-based learning outcomes.
- Instructed a cohort of **50+** students in Data Structures and AI, achieving a **95%** curriculum completion rate and consistently receiving positive feedback for hands-on lab sessions.

KPIT Technologies

Oct 2024 – May 2025

Software Engineering Intern

Pune

- Built a simulation-based AD Learning app to demonstrate ADAS capabilities like Highway Pilot (HP) and Lane Keep Assist (LKA). Leveraged Figma for prototyping and Java/XML for robust application architecture.
- Built a RAG-based backend service using FastAPI and Vector Embeddings to index technical documentation, reducing query resolution time by **30%** and enhancing the learning experience for AD (Autonomous Driving) concepts.
- Engineered and integrated a chatbot using Retrofit for seamless API communication and FastAPI for backend interaction, enhancing user experience with AI-driven scenario-based dialogue.

Publications

- Smart Yield:** Machine Learning-Based Prediction Framework for Optimizing Biodiesel Production. Paper presented at ICCCNT 2025, IIT Indore.

Projects

Predictive Modeling Using Custom Dataset and Ensemble ML

GitHub

- Compiled a custom dataset of 3021 rows from open literature and developed a complete ML regression pipeline using XGBoost, Random Forest, and Gradient Boosting with preprocessing, training, validation, and hyperparameter tuning; also authored a research paper based on this work.
- Built a data cleaning workflow with imputation, outlier removal, multicollinearity reduction, and feature scaling.
- Deployed a Streamlit web app to visualize predictions, evaluation metrics, and model insights interactively.

Agentic RAG System

GitHub

- Engineered an end-to-end Agentic RAG pipeline using LangGraph and a ReAct agent architecture with 2 integrated tools (vector retriever + Wikipedia), reducing manual information lookup time by **70%** and enabling autonomous multi-step reasoning over user-uploaded PDF, TXT, and web documents.
- Built a semantic search engine powered by Voyage-4-Large embeddings and FAISS vector store, processing documents into 500-token chunks with 50-token overlap, achieving sub-2-second retrieval latency across 100+ document chunks.
- Developed a Streamlit web app with real-time PDF/TXT upload, typewriter-style streaming, source citations, and search history, powered by Sarvam-30b LLM and Pydantic-validated state management.

T20 World Cup Analytics & Predictive ML System

GitHub

- Engineered a Python ETL pipeline scraping 1,500+ match scorecards into clean, feature-engineered datasets.
- Trained a Scikit-Learn Random Forest model predicting player impact with **97.5%** accuracy and **0.97 F1-score**.
- Built a Streamlit web dashboard featuring real-time ML inference, dynamic data slicers, and interactive visual analytics.

Skills

Languages: Python, C++, HTML/CSS, JavaScript, SQL

Developer Tools: Linux, VS Code, Git & Github, Power BI, Android Studio, Anaconda Distribution

Technologies/Frameworks: Machine Learning, Deep Learning, cuDA, RAPIDS, Scikit-learn, NumPy, Pandas, PyTorch, TensorFlow, Matplotlib, FastAPI, Streamlit, LangChain, cuDF, cuML, cuPY

Certifications

NVIDIA-Certified Associate: Generative AI and LLMs

[Link](#)

Machine Learning Specialization — Stanford University (Coursera)

[Link](#)